

Long-Term Risk of Dementia After TIA

A Population-Wide Matched Cohort Study

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Abstract

Background and Objectives

Long-term risk of dementia after TIA remains poorly characterized, despite a known impact on cognition. We aimed to quantify the rate, relative risk, and time course of incident dementia after TIA compared with a matched reference population (controls) and individuals with stroke.

Methods

We conducted a population-wide cohort study using linked administrative health data for the province of Ontario, Canada (2002–2022). Adults with a first TIA were identified from emergency department visits or hospital admissions and followed from 90 days after the index event. Individuals with prior dementia, prior stroke, early death, or long-term care admission were excluded. Patients with TIA were matched to population controls on age, sex, neighborhood-level deprivation, rurality, and vascular comorbidities. Patients with TIA were also matched to those with first stroke. Incident dementia was ascertained using a validated algorithm based on hospital diagnoses, physician claims, and dementia-specific medications. Cause-specific and subdistribution hazard models accounting for competing risk of death were used for 1-, 5-, and 10-year risk of dementia, with time-varying analyses.

Results

Among 113,081 matched patients with TIA (mean age 70 years; 50% female), dementia was diagnosed in 19.3% over a mean follow-up of 7.11 years (SD 5.16). Dementia rates after TIA were consistently higher than in matched controls across all time horizons (overall rate 2.71 vs 2.03 per 100 person-years). The hazard ratio for dementia after TIA was highest within the first year (hazard ratio [HR] 1.75, 95% CI 1.66–1.85) and remained elevated over 20 years of follow-up (overall HR 1.34, 95% CI 1.31–1.36). Compared with stroke, the hazard of dementia was lower after TIA early on (1-year HR 0.67, 95% CI 0.64–0.70) but converged and became similar after 5 years of follow-up. Dementia incidence substantially exceeded stroke incidence, and adjustment for incident stroke minimally attenuated results.

Discussion

TIA is associated with a large and sustained increase in dementia risk, approaching that observed after stroke over long-term follow-up. Dementia occurs far more frequently than incident stroke after TIA, highlighting cognitive decline as a major long-term complication and a target for prevention with new strategies that include and extend beyond stroke prevention.

Introduction

Stroke is a well-established risk factor for dementia, and is associated with a risk 2.5x higher than matched controls without stroke.^{1–3} In the 2 decades that follow an initial stroke, a diagnosis of dementia is 3x more common than a recurrent stroke.³ TIAs represent the mildest form of

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Glossary

CIHI = Canadian Institutes for Health Information; **ED** = emergency department; **HR** = hazard ratio; **ICD-10-CA** = International Statistical Classification of Diseases and Related Health Problems, 10th Revision, Canada; **ICES** = Institute for Clinical Evaluative Sciences; **ICH** = intracerebral hemorrhage; **IQR** = interquartile range.

ischemic cerebrovascular disease and account for over 30% of cerebrovascular events.⁴ However, the risk of dementia after a TIA is poorly characterized. Cognitive decline has an accelerated course after TIA,⁵ suggesting the possibility of a higher dementia risk. Although no therapies are currently approved specifically to prevent dementia after TIA or stroke, quantifying the magnitude and timing of dementia risk after TIA, a common diagnosis where secondary prevention is typically less intense than those with stroke,⁶ has important implications for patient counseling, cognitive surveillance, clinical trial design, and prioritization of future mechanistic studies in this population.

In the few cohort studies which have specifically assessed dementia risk in TIA, there have been mixed findings, including a higher risk of dementia after TIA vs controls,^{1,7} no significant difference,⁸ or only an increased risk in those with nonfocal, but not focal symptoms.⁹ There is also no comparison of the time-varying risk of dementia after TIA vs stroke—a population with confirmed ischemic brain injury and high risk of dementia—which would contextualize the magnitude and importance of the risk in the short term and long term. Population-wide linked administrative data could overcome certain biases of inclusion, sample size, and missing data present in prospective cohort studies. As such, we conducted a large population-based analysis of all people diagnosed with TIA in Ontario over a 20-year period, analyzing the rates and temporal pattern of dementia compared with matched controls and matched individuals with stroke.

Methods

Setting and Data Sources

Ontario is Canada's most populous province, with an adult population of over 15 million in 2022. The province provides universal health care with universal access for residents, covering all costs for hospitalizations, emergency department (ED) visits, physician services, and inpatient and some outpatient rehabilitation. However, certain services are not publicly covered, including extended rehabilitation, prescription drug costs for those younger than 65 years, and home care services beyond a certain threshold.

We used the Canadian Institutes for Health Information (CIHI) Discharge Abstract Database to identify hospital admissions and the CIHI-National Ambulatory Care Reporting System to identify ED visits for TIA and acute stroke in Ontario between fiscal years 2002 (starting April 1, 2002) and 2022 (ending March 31, 2022). The Ontario Drug

Benefits Database, the physician claims database, and ICES-derived cohorts (eTables 1 and 2) were used to obtain data on hypertension, diabetes, dyslipidemia, atrial fibrillation, congestive heart failure, coronary artery disease, and peripheral arterial disease using validated algorithms. Other databases are presented in eTable 1. The data sets were linked using unique encoded identifiers and analyzed at ICES (formerly known as the Institute for Clinical Evaluative Sciences).

Participants

Our main cohort comprised individuals aged 18 years and older with first ED visit or hospital admission (most responsible diagnosis) with TIA, identified between 2002 and 2022 in the Ontario population using International Statistical Classification of Diseases and Related Health Problems, 10th Revision, Canada (ICD-10-CA) codes (eTable 1). Those with prior stroke were excluded. Patients with first TIA were not excluded if they had a subsequent stroke to avoid conditioning on the future and creating an unrealistic “pure TIA” cohort that could not have stroke because this would have minimal real-world applicability.¹⁰ We created 2 comparator cohorts. The first was the reference (control) population using the Ontario population during the same period (eFigure 1). The second was a cohort of people with first acute ischemic stroke or intracerebral hemorrhage (ICH) over the same period using ICD-10-CA codes (eTable 1). Codes for TIA and stroke have excellent positive predictive in Canada.¹¹ If there was both a National Ambulatory Care Reporting System (ED visit) and Discharge Abstract Database (hospitalization) code, the index TIA date was taken based on the first presentation in ED. We excluded those with prior stroke, prior dementia, previous residence in long-term care, and those who had death, dementia, or long-term care admission within 90 days from the index event (see eFigure 2).

Matching Process

We first hard matched the TIA cohort to the Ontario reference population. We matched on age (± 1 year), sex, rural/urban residence, neighborhood deprivation index, and comorbidities of hypertension, diabetes, hyperlipidemia, atrial fibrillation, congestive heart failure, coronary artery disease, and peripheral arterial disease (eFigure 3). Only the reference group participants who were alive without dementia at 90 days after the corresponding TIA index date were selected and were assigned the same index date as the matched TIA participant.

Similar matching was performed between the TIA and stroke group. Here, the first event in the period (TIA or stroke) was selected. We used propensity matching instead of hard

matching for comorbidities (eFigure 4) because it led to a higher match rate (propensity matching was not feasible for the TIA-population reference matching because of the lack of a well-defined index date for the population controls [the population controls inherited the index date of the matched TIA patient]). The propensity score was estimated using a logistic regression model in which exposure group (i.e., stroke vs TIA) was regressed on the baseline characteristics. Greedy nearest neighbor matching without replacement was used to match on the logit of the propensity score with a caliper defined as 0.2 of the SD of the logit of the propensity score.¹²

Outcome

We ascertained new diagnoses of dementia using a validated diagnostic algorithm which defines dementia diagnosis as: (1) 1 or more hospitalizations for dementia, OR (2) 1 or more prescriptions for cholinesterase inhibitors (donepezil, galantamine, rivastigmine, tacrine, or memantine), OR (3) 3 or more physician claims with a diagnosis of dementia at least 30 days apart but within a 2-year period (with claims representing the primary reason for the clinical encounter). The diagnostic algorithm has a positive predictive value of 80.4% and negative predictive value of 99%.¹³ Participants were followed for dementia starting 90 days from the date of TIA or stroke. This time point was chosen to exclude patients with early mortality but also any false-positive incident dementia diagnoses in the acute period caused by delirium, ascertainment bias, or by stroke-related language, sensory, or motor barriers to cognitive assessment.^{14,15} Furthermore, this helped to exclude cases of undiagnosed preexisting dementia where cognitive deficits were unmasked or became clinically apparent after an acute event. Cognitive impairment is also dynamic and, in some cases, reversible within the first 90 days; therefore, a diagnosis of dementia may be less reliable during this early period.¹⁶

Statistical Analysis

We compared the baseline characteristics of the matched populations using standardized differences. Standardized differences express the difference between the populations as a proportion of the pooled standard deviation. Unlike traditional hypothesis testing with *p* values, standardized differences are not sensitive to sample size. Standardized differences 0.10 are considered significant.¹⁷ We calculated the crude rates and age/sex-standardized rates (using the 2002 population as standard population) per 100 patient-years of dementia over 1 year, 5 years, 10 years, and total follow-up time, with all participants followed to the first of incident dementia, death, emigration, or September 30, 2022. We did not exclude participants from the reference population who had a subsequent stroke or TIA to avoid bias by comparing to a “healthy” cohort who could not have stroke. Rather, reference group participants were censored when they had an incident TIA, and follow-up for dementia occurred until the time of the incident TIA. If a TIA occurred, that individual was subsequently followed as a participant in the TIA cohort, but we ensured that no individual could be matched to themselves. If a stroke occurred in either the TIA

or reference group, they continued to be followed in the same group so as not to preclude the natural history of both groups. In the stroke-TIA comparison, patients also remained in their respective cohorts after incident stroke.

We used cause-specific hazard models to estimate the hazard ratio (HR) of dementia after TIA compared with the matched groups (TIA vs reference population and TIA vs stroke). These models accounted for the competing risk of death. We allowed the hazard ratio to vary as a smooth function of time using restricted cubic splines, as described elsewhere.¹⁸ For the restricted cubic splines, we selected 3 knots based on equal number of outcome events during the time intervals. We stratified the HR by age and sex and checked interactions by age, sex, urban/rural residence, neighborhood deprivation, and year of study. We also fit subdistribution hazard models (i.e., Fine-Gray models)¹⁹ and estimated cumulative incidence functions. The regression models used robust standard errors to account for the matched nature of the samples.

We conducted some additional sensitivity analyses. First, we evaluated the cumulative incidence of stroke in follow-up and compared it with the cumulative incidence of dementia. We then adjusted for incident stroke to assess whether the association with dementia was driven by incident strokes in follow-up. We adjusted for both (1) incident stroke as a binary event within 90 days of TIA (before dementia follow-up) and (2) time-varying stroke after 90 days (during follow-up). We assessed any change to the cause-specific dementia HR for the TIA reference group after adjusting for incident stroke. Second, given the potential for TIA to represent an episode related to cognitive decline or delirium, we repeated the analysis starting follow-up 1 year after TIA to exclude dementia diagnosed within that year. Third, some TIA diagnoses may represent transient focal neurologic episodes from cerebral amyloid angiopathy, so we assessed the future rate of ICH and subarachnoid hemorrhage to see to what degree they accounted for the differences in dementia rates. Fourth, we report the dementia HR among those seen by a neurologist during their hospitalization. Fifth, we report the dementia HR after excluding those with an outpatient diagnosis of stroke within 30 days of their ED visit, to mitigate the potential of the index TIA diagnosis representing a minor stroke.

Standard Protocol Approvals, Registrations, and Patient Consents

ICES is an independent, nonprofit research institute whose legal status under Ontario’s health information privacy law allows it to collect and analyze health care and demographic data, without consent, for health system evaluation and improvement. The use of the data in this project is authorized under section 45 of Ontario’s Personal Health Information Protection Act and does not require review by a Research Ethics Board.

Data Availability

The data set from this study is held securely in coded form at ICES. Although legal data sharing agreements between ICES

and data providers (e.g., health care organizations and government) prohibit ICES from making the data set publicly available, access may be granted to those who meet pre-specified criteria for confidential access, available at ices.on.ca/DAS (email: das@ices.on.ca). The full data set creation plan and underlying analytic code are available from the authors on request, understanding that the computer programs may rely on coding templates or macros that are unique to ICES and are therefore either inaccessible or may require modification.

Results

There were 113,780 patients with TIA in the unmatched cohort after applying exclusion criteria (eFigure 2). A total of 113,081 people with TIA were successfully matched to the reference population without TIA at baseline (99.4% match). The 2 groups were perfectly matched, with mean age of the matched cohorts 70.0 years (SD 13.4), 50.3% female, 16.8% residing in rural areas, 26.4% with diabetes, 69.3% with hypertension, 16.7% with atrial fibrillation, 10.8% with congestive heart failure, 13.1% with coronary artery disease, and 3.6% with peripheral artery disease (Table 1).

Comparison Between TIA Cohort and Reference Population

Overall (total) follow-up for the TIA group was a mean of 7.11 years (SD 5.16), median 5.90 (interquartile range [IQR] 2.90–10.40), and range 0–21.3 years, and for the reference group was a mean follow-up of 7.65 years (SD 5.23), median of 6.50 (IQR 3.30–11.1), and range of 0–21.3 years. Between 90 days after stroke and end of follow-up, dementia was diagnosed in 21,818 (19.3%) with TIA compared with 17,537 (15.2%) of the matched reference population. Dementia outpatient claims were most commonly made by family physicians/general practitioners (66.3%), followed by geriatricians (17.9%), internists (5.7%), neurologists (4.9%), psychiatrists (3.9%), and other specialties (1.3%). Mortality was slightly higher after TIA (N = 32,720; 28.9%) compared with reference population (N = 29,984; 26.5%).

The rate of dementia within 1 year was 3.19 per 100 patient-years (95% CI 3.08–3.29) in the TIA cohort compared with 1.82 (1.74–1.90) in the reference population (Table 2). The rates of dementia after TIA were higher compared with the reference population within 5 years (2.77 per 100 patient-years, 95% CI 2.72–2.82 vs 1.90 per 100 patient-years, 95% CI 1.86–1.94), within 10 years (2.73, 95% CI 2.69–2.77 vs 1.97, 95% CI 1.94–2.01), and with total follow-up (2.71, 95% CI

Table 1 Baseline Characteristics of TIA Cohorts Matched With Ontario Reference Population and With Stroke Cohort

Baseline factor	TIA group (N = 113,081)	Reference group (N = 113,081)	SD	TIA (N = 100,048)	Stroke (N = 100,048)	SD
Age groups, n (%)						
Age 18–44	4,586 (4.1)	4,589 (4.1)	0	3,793 (3.8)	3,793 (3.8)	0
Age 45–64	31,670 (28.0)	31,678 (28.0)	0	28,482 (28.5)	28,482 (28.5)	0
Age 65–74	29,509 (26.1)	29,503 (26.1)	0	25,978 (26.0)	25,978 (26.0)	0
Age 75–84	32,436 (28.7)	32,391 (28.6)	0.001	29,048 (29.0)	29,048 (29.0)	0
Age 85+	14,880 (13.2)	14,920 (13.2)	0.001	12,747 (12.7)	12,747 (12.7)	0
Age, mean (SD)	70.0 (13.4)	70.0 (13.4)	0.001	69.9 (13.3)	70.1 (13.3)	0.014
Female, n (%)	56,184 (49.7)	56,184 (49.7)	0	50,692 (50.7)	50,692 (50.7)	0
Rural, n (%)	18,949 (16.8)	18,949 (16.8)	0	13,724 (13.7)	13,724 (13.7)	0
Highest material deprivation quintile, n (%)	22,731 (20.1)	22,731 (20.1)	0	21,835 (21.8)	21,835 (21.8)	0
Diabetes, n (%)	29,810 (26.4)	29,810 (26.4)	0	26,645 (26.6)	27,019 (27.0)	0.008
Hyperlipidemia, n (%)	40,468 (35.8)	40,468 (35.8)	0	34,656 (34.6)	37,147 (37.1)	0.052
Hypertension, n (%)	78,386 (69.3)	78,386 (69.3)	0	72,194 (72.2)	71,591 (71.6)	0.013
Atrial fibrillation, n (%)	18,895 (16.7)	18,895 (16.7)	0	17,318 (17.3)	16,984 (17.0)	0.009
CHF, n (%)	12,173 (10.8)	12,173 (10.8)	0	10,710 (10.7)	10,718 (10.7)	0
CAD, n (%)	14,810 (13.1)	14,810 (13.1)	0	12,031 (12.0)	13,418 (13.4)	0.042
PVD, n (%)	4,118 (3.6)	4,118 (3.6)	0	3,721 (3.7)	3,824 (3.8)	0.005

Abbreviations: CAD = coronary artery disease; CHF = congestive heart failure; PVD = peripheral vascular disease; SD = standardized difference.

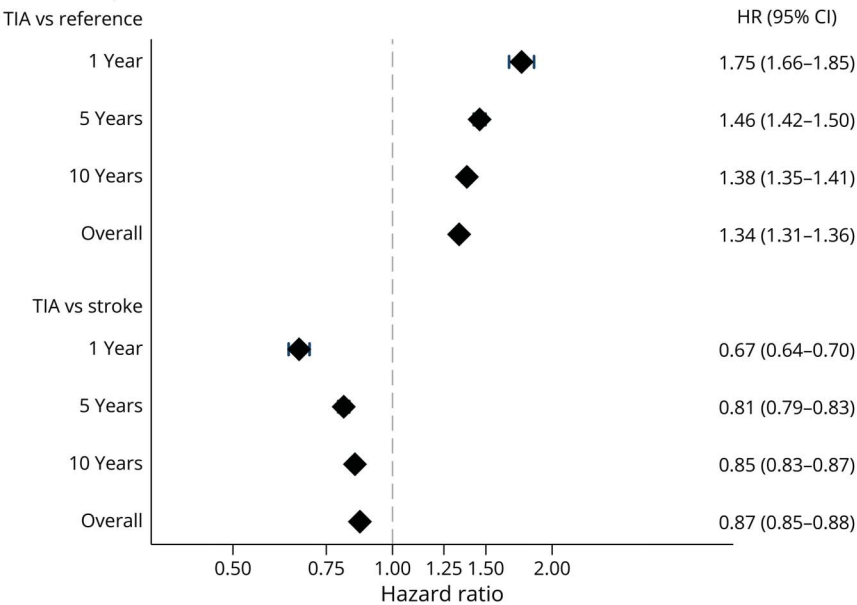
Table 2 Rate of Dementia per 100 Patient-Years in People With TIA and Ontario Reference Population, Overall and Stratified by Age and Sex

Group	Events in 1-y, n	Rate per 100 person-year (95% CI) in 1-y	Events in 5-y, n	Rate per 100 person-year (95% CI) in 5-y	Events in 10-y, n	Rate per 100 person-year (95% CI) in 10-y	Events overall, n	Rate per 100 person-year (95% CI) at overall
Overall: TIA	3,469/113,081	3.19 (3.08–3.29)	12,341/113,081	2.77 (2.72–2.82)	18,390/113,081	2.73 (2.69–2.77)	21,818/113,081	2.71 (2.68–2.75)
Overall: reference	2,007/113,081	1.82 (1.74–1.90)	8,783/113,081	1.90 (1.86–1.94)	14,094/113,081	1.97 (1.94–2.01)	17,537/113,081	2.03 (2.00–2.06)
Age 18–44: TIA	<6/4,589 ^a	0.02 (0.00–0.16)	9/4,589	0.04 (0.02–0.08)	17/4,589	0.05 (0.03–0.08)	31/4,589	0.07 (0.05–0.09)
Age 18–44: reference	<6/4,586 ^a	0.00 (0.00–0.00)	<6/4,586 ^a	0.00 (0.00–0.03)	<6/4,586 ^a	0.01 (0.01–0.03)	14/4,586	0.03 (0.02–0.05)
Age 45–64: TIA	141/31,678	0.45 (0.38–0.53)	547/31,678	0.39 (0.36–0.42)	1,066/31,678	0.46 (0.43–0.49)	1,774/31,678	0.60 (0.57–0.62)
Age 45–64: reference	39/31,670	0.12 (0.09–0.17)	231/31,670	0.16 (0.14–0.18)	531/31,670	0.22 (0.20–0.24)	1,079/31,670	0.35 (0.33–0.37)
Age 65–74: TIA	489/29,503	1.70 (1.56–1.86)	2,146/29,503	1.76 (1.69–1.84)	3,815/29,503	2.03 (1.97–2.10)	5,221/29,503	2.34 (2.28–2.40)
Age 65–74: reference	177/29,509	0.61 (0.52–0.70)	1,065/29,509	0.84 (0.79–0.89)	2,359/29,509	1.18 (1.13–1.22)	3,716/29,509	1.53 (1.48–1.58)
Age 75–84: TIA	1,551/32,391	5.04 (4.80–5.30)	5,780/32,391	4.86 (4.74–4.99)	8,728/32,391	5.22 (5.12–5.33)	9,916/32,391	5.45 (5.35–5.56)
Age 75–84: reference	870/32,436	2.77 (2.60–2.96)	4,149/32,436	3.30 (3.20–3.40)	6,929/32,436	3.81 (3.72–3.90)	8,298/32,436	4.12 (4.04–4.21)
Age 85+: TIA	1,287/14,920	9.55 (9.05–10.09)	3,859/14,920	8.80 (8.53–9.08)	4,764/14,920	9.01 (8.76–9.27)	4,876/14,920	9.07 (8.82–9.33)
Age 85+: reference	921/14,880	6.69 (6.27–7.13)	3,337/14,880	7.20 (6.96–7.45)	4,270/14,880	7.47 (7.25–7.70)	4,430/14,880	7.56 (7.34–7.79)
Female: TIA	1,928/56,897	3.52 (3.37–3.68)	6,870/56,897	3.08 (3.00–3.15)	10,266/56,897	3.03 (2.97–3.09)	12,181/56,897	3.02 (2.97–3.07)
Female: reference	1,168/56,897	2.10 (1.99–2.23)	5,229/56,897	2.25 (2.19–2.31)	8,288/56,897	2.31 (2.26–2.36)	10,278/56,897	2.37 (2.33–2.42)
Male: TIA	1,541/56,184	2.85 (2.71–2.99)	5,471/56,184	2.46 (2.40–2.53)	8,124/56,184	2.42 (2.37–2.47)	9,637/56,184	2.41 (2.36–2.46)
Male: reference	839/56,184	1.53 (1.43–1.64)	3,554/56,184	1.54 (1.49–1.59)	5,806/56,184	1.63 (1.59–1.68)	7,259/56,184	1.69 (1.65–1.73)

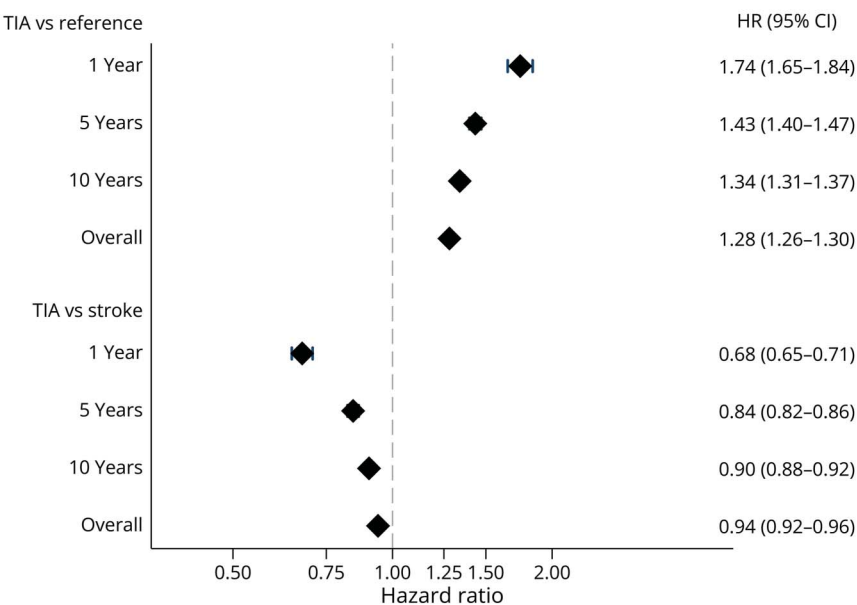
Abbreviation: ICES = Institute for Clinical Evaluative Sciences.
^aExact number suppressed due to small cell as per ICES policy.

Figure 1 HRs Comparing Matched Groups

A. Cause-specific hazard ratios



B. Sub-distribution hazard ratios



There is a higher dementia HR TIA compared with reference population and lower dementia HR when comparing TIA with stroke, when evaluating (A) cause-specific or (B) subdistribution HRs. HRs = hazard ratios.

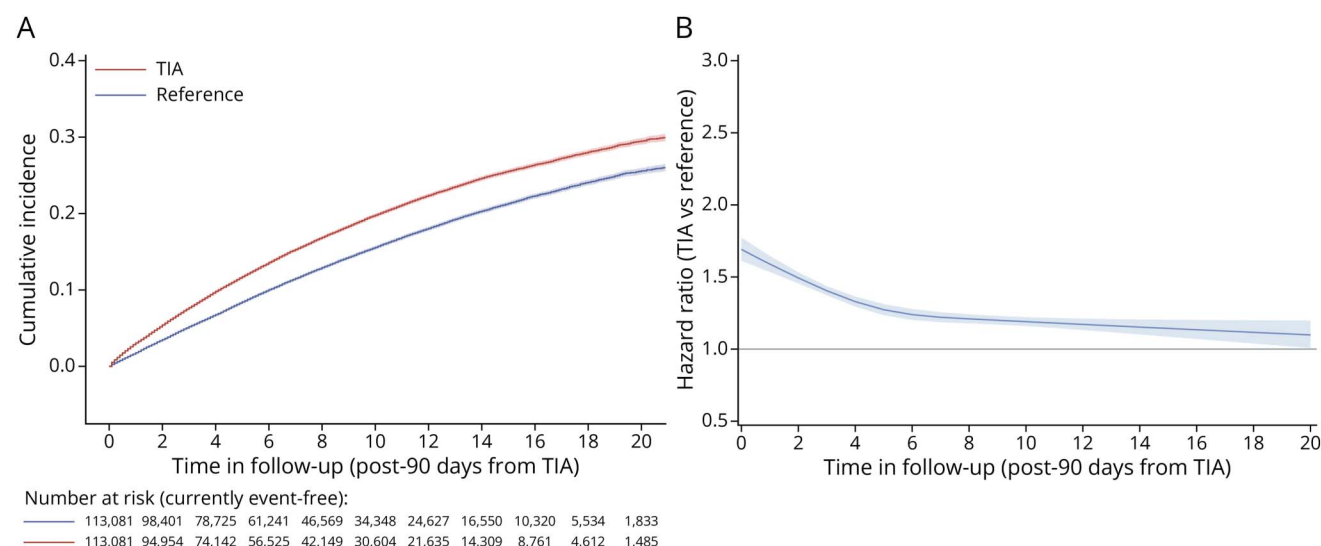
2.68–2.75 vs 2.03 95% CI 2.00–2.06; Table 2). There was a consistently greater rate of dementia for those with TIA compared with reference population in each age strata and in women and men (Table 2).

The cause-specific HR for dementia after TIA within 1 year was 1.75 (95% CI 1.66–1.85), within 5 years was 1.46 (95% CI 1.42–1.50), within 10 years was 1.38 (95% CI 1.35–1.41), and for total follow-up was 1.34 (95% CI 1.31–1.36; Figure 1A). Direction of HRs was similar with Fine-Gray models (Figure 1B). The absolute rates of post-TIA dementia were highest with older age and in women (Table 2), but

there was a greater relative impact of TIA on dementia (higher HR) in younger people compared with older people (interaction p -value <0.001) and in men compared with women (interaction p -value <0.001; eTable 3). There was no interaction of TIA-dementia association by urban/rural residence, neighborhood deprivation, or year of study (eTable 3).

Comparison Between Index TIA and Index Stroke

A total of 100,048 people with TIA were matched to people with stroke (90.9% match), and the 2 groups were well matched (Table 1). Over total follow-up, dementia was

Figure 2 Time Course of Dementia Risk After TIA

(A) Cumulative incidence of dementia and (B) time-varying hazard ratios comparing TIA with reference population.

diagnosed in 19,372 (19.4%) in the TIA cohort compared with 20,099 (20.1%) in the stroke cohort. The rate of dementia at 1 year was 3.17 (95% CI 3.06–3.28) per 100 patient-years in the TIA cohort compared with 4.77 (4.63–4.91) after stroke and slightly lower after TIA compared with stroke at 5, 10 years, and total follow-up, with the relative difference decreasing with longer follow-up time (eTable 4). The HR of dementia (Figure 1, A and B) was lower after TIA compared with stroke, but the HR was less pronounced with longer follow-up (HR within 1 year 0.67, 95% CI 0.64–0.70, HR within 5 years 0.81, 95% CI 0.79–0.83, HR within 10 years 0.85, 95% CI 0.83–0.87).

Time Course of Dementia Diagnosis

There was continuous accrual of dementia events starting from 3 months after TIA until the end of follow-up time, and this was higher after TIA than in the matched reference population throughout the entire follow-up time (Figure 2A). Time-varying HRs were markedly higher within the first year after TIA compared with the reference population and decreased over time but remained elevated even 20 years after TIA (Figure 2B).

By contrast, cumulative incidence of dementia was slightly lower after TIA compared with stroke, but the gap narrowed, and cumulative incidence became similar after 16 years of follow-up (Figure 3A). Time-varying hazard ratios for dementia comparing TIA to stroke showed an initially smaller HR under 0.7 which rapidly increased to near 1 within 5 years, implying a similar rate of new dementia accrual thereafter (Figure 3B).

Incident Stroke

Incident stroke in the follow-up period occurred in 7.67% of those with index TIA and 10.1% of those with index stroke.

However, in those with either index TIA or stroke, the cumulative incidence of dementia was far higher than the cumulative incidence of incident stroke in follow-up (Figure 3C).

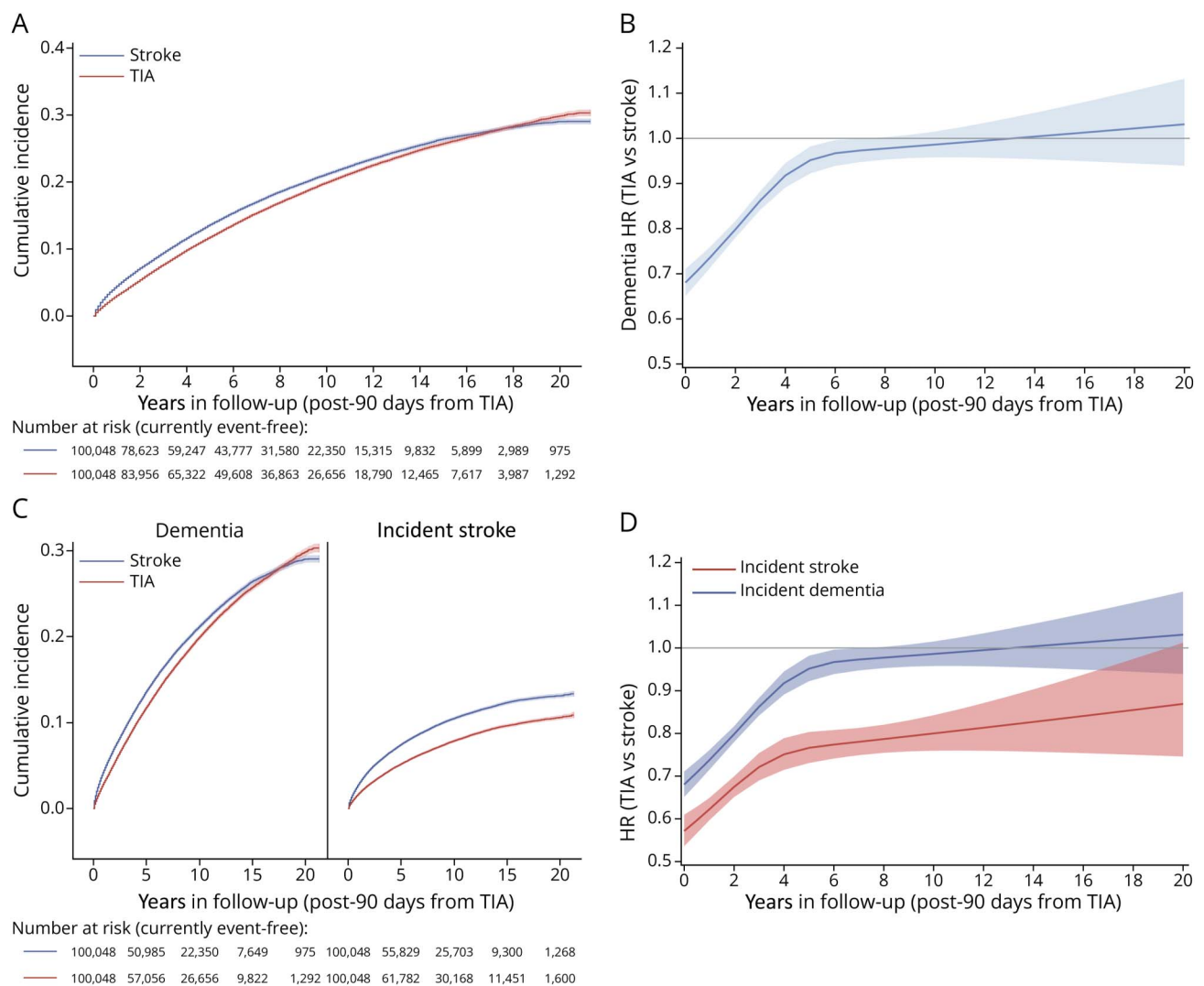
The lower risk of incident stroke among individuals with index TIA compared with index stroke was pronounced and persistent across the entire follow-up period. By contrast, the lower risk of incident dementia after TIA compared with stroke was less pronounced, and risks were similar after approximately 5 years (Figure 3, C and D).

Sensitivity Analyses

After TIA, the occurrence of stroke was associated with higher risk of dementia, whether within 90 days of TIA (HR 1.91, 95% CI 1.48–2.46) or after 90 days (HR 2.11, 95% CI 1.96–2.27). After adjustment for occurrence of incident stroke the HR of dementia was only slightly attenuated for TIA vs reference population (HR 1.26, 95% CI 1.24–1.29, compared with HR 1.34 before adjustment) and for TIA vs index stroke (HR 0.89, 95% CI 0.87–0.90, compared with HR 0.87). The results were similar in a 1-year landmark analysis (eTable 5).

When separating into stroke types, the risk of both ischemic stroke and ICH was elevated after TIA compared with the reference population; however, the rates of ICH and SAH were small and far less than those of dementia (eFigure 5).

We found that 14,299 people with TIA (12.6%) were seen by a neurologist during their hospitalization. The HR of dementia including only those with TIA seen by a neurologist and their matched control was slightly higher (1.44, 95% CI

Figure 3 Time Course of Dementia Comparing People With Index TIA and Index Stroke

(A) Cumulative incidence of dementia, (B) time-varying hazard ratio (HR), (C) cumulative incidence of both dementia and incident stroke, and (D) time-varying hazard ratio of dementia and incident stroke.

1.37–1.5) compared with the overall TIA population (1.34, 95% CI 1.31–1.36).

After excluding people with an outpatient diagnostic code for stroke within 30 days after TIA, we were left with 84,584 matched pairs (75% of original cohort) which yielded a similar HR for dementia (HR 1.32, 95% CI 1.29–1.35).

Discussion

We had 3 main findings in this large, population-wide study of dementia risk after TIA. First, we demonstrated an increased hazard of dementia after TIA compared with matched controls of 34% over a mean follow-up of 7 years, and the risk remained elevated over 20 years of follow-up, despite matching for neighborhood deprivation, age, sex, and vascular risk factors; adjusting for incident stroke; and accounting for

the competing risk of death. This increased risk was 75% higher in the first year after TIA. Second, the long-term incidence of dementia after TIA and stroke was comparable. Finally, our results show a 3x higher rate of dementia compared with incident stroke, highlighting dementia as a major long-term complication after TIA.

Using a whole population approach that allowed a vastly larger sample size and number of dementia events than previous studies, we were able to report estimates with greater precision and add evidence to prior discrepancies in the TIA dementia risk association. The risk of dementia after stroke is well established from large epidemiologic studies,^{1–3} and there are few data on TIA alone.^{1–3,20} As such, the importance of dedicated longitudinal studies of TIA and cognitive outcomes has been raised.²¹ TIA has previously been shown to increase the long-term risk of disability, myocardial infarction, and mortality.²² Establishing TIA as a risk factor for dementia

improves risk stratification, highlights the importance of monitoring for cognitive decline, and informs inclusion of patients with TIA in prospective cohorts, biomarker studies, and clinical trials aimed at reducing dementia incidence.

In clinical practice, those with a cerebrovascular event that resolves within 24 hours and is not accompanied by brain imaging evidence of stroke are commonly diagnosed with TIA. Given our study design, we did not have access to imaging data, particularly magnetic resonance imaging (MRI). Despite a shift toward use of MRI to rule out minor stroke,²³ most patients worldwide do not receive MRI, and the proportion of patients diagnosed with TIA remains high at approximately 30% of all cerebrovascular events.⁴ Although there are proposals to cease distinguishing between TIA and minor stroke,²⁴ MRI is not universally performed or available. Requiring MRI for the diagnosis of TIA/minor stroke may further entrench socioeconomic disparities because individuals with higher socioeconomic status are more likely to receive MRI, even within a universal health care system.²⁵ As such, TIA remains a common and clinically useful syndrome to identify patients with transient neurologic deficits who remain at high risk of subsequent stroke and cardiovascular events. However, patients with TIA or minor stroke may receive limited follow-up due to resource constraints, the perception of a transient event, and limited recognition of residual impairments,²⁶ and therefore long-term outcomes, including cognitive decline, may be underrecognized.

Our findings have important implications for understanding dementia after TIA. Mechanistic frameworks for poststroke dementia typically assume a degree of brain injury that causes direct cognitive impairment or accelerates or unmasks neurodegenerative processes. By contrast, patients with TIA are commonly reassured that there is no permanent brain injury or that any injury is relatively small in the case of occult minor stroke. Nevertheless, we observed that the long-term risk of dementia after TIA approaches that seen after stroke, indicating that mechanisms underlying cognitive decline in this population require reconsideration. Although incident overt or covert stroke are immediate possibilities, adjustment for incident stroke only reduced the hazard ratio by about a third and it remained significant. Notably, dementia occurred at a rate approximately 3 times higher than incident stroke, underscoring dementia as a major long-term complication rather than a downstream consequence of incident cerebrovascular events alone. Furthermore, although TIA and stroke remained distinct with respect to future stroke risk, the long-term risk of dementia became increasingly similar. While prevention of early stroke after TIA has appropriately been the primary clinical focus, secondary prevention efforts are often less intense than those with stroke,⁶ and our findings revalidate the importance of post-TIA stroke prevention given the doubled risk of dementia after subsequent stroke. Our findings also demonstrate that dementia risk remains elevated throughout follow-up, particularly within the first 5 years, with

long-term rates similar to post-stroke dementia. Prior work suggests that after TIA, there is no abrupt cognitive loss but rather a steeper downward trajectory, supporting the role of secondary processes such as accelerated neurodegeneration, neuroinflammation, endothelial dysfunction, blood-brain barrier disruption, or microischemia.²⁷⁻³¹ Demonstrating a substantial burden of dementia after TIA justifies ongoing efforts to identify modifiable vascular and nonvascular mechanisms contributing to cognitive decline. However, it should be acknowledged that other unrecognized drivers causing both TIA and dementia are possible. These findings support consideration of cognitive outcomes alongside incident stroke as major long-term outcomes after TIA and suggest that dementia prevention after TIA will require strategies beyond stroke prevention alone.

There are limitations to the study. First, we relied on diagnostic codes for TIA, and the diagnosis of TIA is challenging in practice because it needs to be separated from mimics such as seizure, migraine, transient global amnesia, other nonspecific symptoms, or delirium, particularly that related to previously undiagnosed cognitive decline. However, this was a design decision to include a large and representative population and to reflect the reality of heterogeneous TIA diagnosis in clinical practice. For example, posterior circulation symptoms present a particular clinical challenge, yet studies suggest that transient neurologic symptoms, particularly isolated brain stem symptoms that do not fit the conventional diagnostic criteria for TIA may be associated with posterior circulation strokes, coronary events, and dementia.^{9,32,33} In a study of transient neurologic events preceding stroke, only 8% of nonspecific transient neurologic attacks preceding vertebrobasilar stroke (suggesting that these attacks had ischemic origin) fulfilled National Institute of Neurologic Disorders and Stroke criteria for TIA.³³ Furthermore, TIA diagnostic codes have shown high specificity and PPV, and although the sensitivity of the TIA codes is modest,^{11,34,35} the large number of participants in the study would likely have captured a much broader population than previous cohort studies. Furthermore, in 1 study, those with mixed or nonfocal symptoms, who may be less likely to be coded as TIA, had a higher risk of stroke and dementia than those with focal symptoms (and therefore, lower sensitivity for more nonspecific TIAs may bias our results toward the null).⁹ Finally, improving diagnostic accuracy for TIA would require all participants be seen by stroke trained neurologists, which is not feasible for a population-based study, and furthermore, the agreement between stroke-trained neurologists on TIA was poor.³⁶ We found that among those seen by a neurologist during their hospitalization, the HR for dementia was only slightly higher. The HR for dementia was also similar when excluding those with an outpatient physician claim for stroke within 1 month, increasing the robustness of our findings. It should also be noted that despite the differences in the approach, our high rates of dementia after TIA were concordant with the OXVASC study which has detailed characterization of index events.¹ In addition, the hazard ratio

for dementia was consistent across years of enrollment, during which increase in MRI use occurred.³⁷ Second, we did not have detailed brain imaging to assess covert disease at baseline or follow-up. Lacunes and early confluent or confluent white matter hyperintensities are more common in participants with TIA.³⁸ However, our goal was to establish the risk of dementia after TIA, regardless of underlying cause, and further detailed prospective cohort studies will be required to identify the mechanisms. Third, we did not have cognitive testing and could not determine whether the TIA symptoms were cognitive in nature or led to additional cognitive testing that revealed preexisting cognitive impairment or dementia. However, we excluded the first 3 months of follow-up to avoid immediate post-TIA dementia diagnoses which are more likely to reflect recognition of pre-TIA cognitive decline also found that dementia continued to accrue over time rather than being purely frontloaded. We also found similar results after excluding the first year (further excluding early diagnoses of dementia that may have reflected pre-TIA cognitive decline) and found that the rates of ICH and SAH (markers of cerebral amyloid angiopathy) were very low and could not explain the dementia differences. Fourth, our follow-up relied on dementia coding rather than detailed clinical assessments; however, the use of diagnostic codes for both exposure and outcome enables population-level inference that would not otherwise be possible. The dementia definition is validated and has a high specificity but a sensitivity of 80%, likely missing some people with mild dementia; however, this would have the impact of biasing our results toward the null. We could not determine the extent of the assessments and investigations, but the diagnostic algorithm reflected real-world diagnostic patterns in a publicly funded health care system where general practitioner and geriatric assessments were most common. We could not determine dementia subtype, that is, Alzheimer disease or vascular dementia, because the algorithm was validated for all-cause dementia. Because cerebrovascular disease can cause vascular dementia or accelerate other underlying neuropathologies and is often associated with mixed (vascular-neurodegenerative) dementia,³⁰ we believe all-cause dementia best captures the totality of the association of TIA with dementia in this large cohort. Our population-based results should complement more deeply phenotyped cohort studies incorporating neuroimaging and biomarkers.

In conclusion, this large population-level study demonstrates that 1 in 5 patients with TIA develop dementia over a mean of 7 years of follow-up, with a 34% higher hazard than controls, and a long-term risk approaching that seen after stroke. Dementia is a common secondary complication after TIA and, together with poststroke dementia, represents an important target for future research and clinical trials to lower the risk.

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Author Contributions

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References

- Pendlebury ST, Rothwell PM; Oxford Vascular Study. Incidence and prevalence of dementia associated with transient ischaemic attack and stroke: analysis of the population-based Oxford Vascular Study. *Lancet Neurol*. 2019;18(3):248-258. doi:10.1016/S1474-4422(18)30442-3
- Corraini P, Henderson VW, Oding AG, Pedersen L, Horváth-Puhó E, Sørensen HT. Long-term risk of dementia among survivors of ischemic or hemorrhagic stroke. *Stroke*. 2017;48(1):180-186. doi:10.1161/STROKEAHA.116.015242
- Joundi RA, Fang J, Austin PC, et al. Magnitude and time-course of dementia risk in stroke survivors: a population-wide matched cohort study. *Neurology*. 2025;104(8):e213478. doi:10.1212/WNL.000000000000213478
- von Weitzel-Mudersbach P, Andersen G, Hundborg HH, Johnsen SP. Transient ischemic attack and minor stroke are the most common manifestations of acute cerebrovascular disease: a prospective, population-based study: the Aarhus TIA study. *Neuroepidemiology*. 2013;40(1):50-55. doi:10.1159/000341696
- Del Bene VA, Howard G, Groppen TI, et al. Cognitive decline after first-time transient ischemic attack. *JAMA Neurol*. 2025;82(4):323-332. doi:10.1001/jamaneurol.2024.5082
- Fleet JL, Carter B, Kapral MK, Lam M, Frisbee S, Shariff SZ. Post TIA care in Ontario: a population-based cohort study on secondary prevention. *J Neurol Sci*. 2025;476:123603. doi:10.1016/j.jns.2025.123603
- Lakshminarayan K, Norby FL, Hong C-P, et al. Stroke, cardiovascular disease, and dementia after hospitalized TIA: the Atherosclerosis Risk in Communities Study. *Neurology*. 2025;105(6):e214020. doi:10.1212/WNL.000000000000214020
- Lioutas V, Peloso G, Romero JR, et al. Long-term incidence of dementia following transient ischemic attack: a longitudinal cohort study. *J Am Heart Assoc*. 2025;14:e037817. doi:10.1161/JAHA.124.037817
- Bos MJ, van Rijn MJE, Witterman JCM, Hofman A, Koudstaal PJ, Breteler MMB. Incidence and prognosis of transient neurological attacks. *JAMA*. 2007;298(24):2877-2885. doi:10.1001/jama.298.24.2877
- Karim ME, Gustafson P, Petkau J, Tremlett H; Long-Term Benefits and Adverse Effects of Beta-Interferon for Multiple Sclerosis (BeAMS) Study Group. Comparison of statistical approaches for dealing with immortal time bias in drug effectiveness studies. *Am J Epidemiol*. 2016;184(4):325-335. doi:10.1093/aje/kwv445
- Porter J, Mondor L, Kapral MK, Fang J, Hall RE. How reliable are administrative data for capturing stroke patients and their care. *Cerebrovasc Dis Extra*. 2016;6(3):96-106. doi:10.1159/000449288
- Austin PC, Yu AYY, Vyas MV, Kapral MK. Applying propensity score methods in clinical research in neurology. *Neurology*. 2021;97(18):856-863. doi:10.1212/WNL.00000000000012777
- Jaakkimainen RL, Bronskill SE, Tierney MC, et al. Identification of physician-diagnosed Alzheimer's disease and related dementias in population-based administrative data: a validation study using family physicians' electronic medical records. *J Alzheimers Dis*. 2016;54(1):337-349. doi:10.3233/JAD-160105
- Pendlebury ST, Klaus SP, Thomson RJ, et al. Methodological factors in determining risk of dementia after transient ischemic attack and stroke: (III) applicability of cognitive tests. *Stroke*. 2015;46(11):3067-3073. doi:10.1161/STROKEAHA.115.010290
- Román GC, Tatemichi TK, Erkinjuntti T, et al. Vascular dementia: diagnostic criteria for research studies. Report of the NINDS-AIREN International Workshop. *Neurology*. 1993;43(2):250-260. doi:10.1212/wnl.43.2.250
- Sivakumar L, Kate M, Jeerakathil T, Camicioli R, Buck B, Butcher K. Serial Montreal cognitive assessments demonstrate reversible cognitive impairment in patients with acute transient ischemic attack and minor stroke. *Stroke*. 2014;45(6):1709-1715. doi:10.1161/STROKEAHA.114.004726
- Austin PC. Using the standardized difference to compare the prevalence of a binary variable between two groups in observational research. *Commun Stat Simul Comput*. 2009;38(6):1228-1234. doi:10.1080/03610910902859574
- Austin PC, Fang J, Lee DS. Using fractional polynomials and restricted cubic splines to model non-proportional hazards or time-varying covariate effects in the Cox regression model. *Stat Med*. 2022;41(3):612-624. doi:10.1002/sim.9259
- Austin PC, Lee DS, Fine JP. Introduction to the analysis of survival data in the presence of competing risks. *Circulation*. 2016;133(6):601-609. doi:10.1161/CIRCULATIONAHA.115.017719
- Koton S, Pike JR, Johansen M, et al. Association of ischemic stroke incidence, severity, and recurrence with dementia in the Atherosclerosis Risk in Communities Cohort Study. *JAMA Neurol*. 2022;79(3):271-280. doi:10.1001/jamaneurol.2021.5080
- Hayes A, Kasner SE, Favilla CG, et al. Not So transient? A narrative review on cognitive impairment after transient ischemic attack. *Stroke*. 2024;55(10):2558-2566. doi:10.1161/STROKEAHA.124.046821
- Writing Committee for the PERSIST Collaborators, Khan F, Yogendrakumar V, et al. Long-term risk of stroke after transient ischemic attack or minor stroke: a systematic review and meta-analysis. *JAMA*. 2025;333(17):1508-1519. doi:10.1001/jama.2025.2033
- Sacco RL, Kasner SE, Broderick JP, et al. An updated definition of stroke for the 21st century. *Stroke*. 2013;44(7):2064-2089. doi:10.1161/str.0b013e318296aeca
- Easton JD, Johnston SC. Time to retire the concept of transient ischemic attack. *JAMA*. 2022;327(9):813-814. doi:10.1001/jama.2022.0300
- You JJ, Venkatesh V, Laupacis A. Better access to outpatient magnetic resonance imaging in Ontario: but for whom? *Open Med*. 2009;3(1):e22-e25.
- Turner GM, Aquino MRJV, Atkins L, Foy R, Mant J, Calvert M. Factors influencing follow-up care post-TIA and minor stroke: a qualitative study using the theoretical domains framework. *BMC Health Serv Res*. 2022;22(1):235. doi:10.1186/s12913-022-07607-0
- Mok VCT, Cai Y, Markus HS. Vascular cognitive impairment and dementia: mechanisms, treatment, and future directions. *Int J Stroke*. 2024;19(8):838-856. doi:10.1177/17474930241279888
- Mok VCT, Lam BYK, Wong A, Ko H, Markus HS, Wong LKS. Early-onset and delayed-onset poststroke dementia: revisiting the mechanisms. *Nat Rev Neurol*. 2017;13(3):148-159. doi:10.1038/nrneurol.2017.16
- Rost NS, Brodtmann A, Pase MP, et al. Post-stroke cognitive impairment and dementia. *Circ Res*. 2022;130(8):1252-1271. doi:10.1161/CIRCRESAHA.122.319951
- Ganesh A, Barber PA. The cognitive sequelae of transient ischemic attacks-recent insights and future directions. *J Clin Med*. 2022;11(9):2637. doi:10.3390/jcm11092637
- Munir M, Urtenbach J, Reid M, et al. Longitudinal brain atrophy rates in transient ischemic attack and minor ischemic stroke patients and cognitive profiles. *Front Neurol*. 2019;10:18. doi:10.3389/fneur.2019.00018
- Koudstaal PJ, Algra A, Pop GA, Kappelle LJ, van Latum JC, van Gijn J. Risk of cardiac events in atypical transient ischaemic attack or minor stroke. The Dutch TIA Study Group. *Lancet*. 1992;340(8820):630-633. doi:10.1016/0140-6736(92)92170-k
- Paul NLM, Simoni M, Rothwell PM; Oxford Vascular Study. Transient isolated brainstem symptoms preceding posterior circulation stroke: a population-based study. *Lancet Neurol*. 2013;12(1):65-71. doi:10.1016/S1474-4422(12)70299-5
- Yu AYY, Quan H, McRae A, Wagner GO, Hill MD, Coutts SB. Moderate sensitivity and high specificity of emergency department administrative data for transient ischemic attacks. *BMC Health Serv Res*. 2017;17(1):666. doi:10.1186/s12913-017-2612-6
- Hall R, Mondor L, Porter J, Fang J, Kapral MK. Accuracy of administrative data for the coding of acute stroke and TIAs. *Can J Neurol Sci*. 2016;43(6):765-773. doi:10.1017/cjn.2016.278
- Castle J, Mlynash M, Lee K, et al. Agreement regarding diagnosis of transient ischemic attack fairly low among stroke-trained neurologists. *Stroke*. 2010;41(7):1367-1370. doi:10.1161/STROKEAHA.109.577650
- Ng V, Bayoumi A, Fang J, et al. Temporal trends in the use of investigations after stroke or transient ischemic attack. *Med Care*. 2016;54(5):430-434. doi:10.1097/MLR.0000000000000499
- Oudeman EA, Greving JP, Van den Berg-Vos RM, et al. Nonfocal transient neurological attacks are associated with cerebral small vessel disease. *Stroke*. 2019;50(12):3540-3544. doi:10.1161/STROKEAHA.119.025328